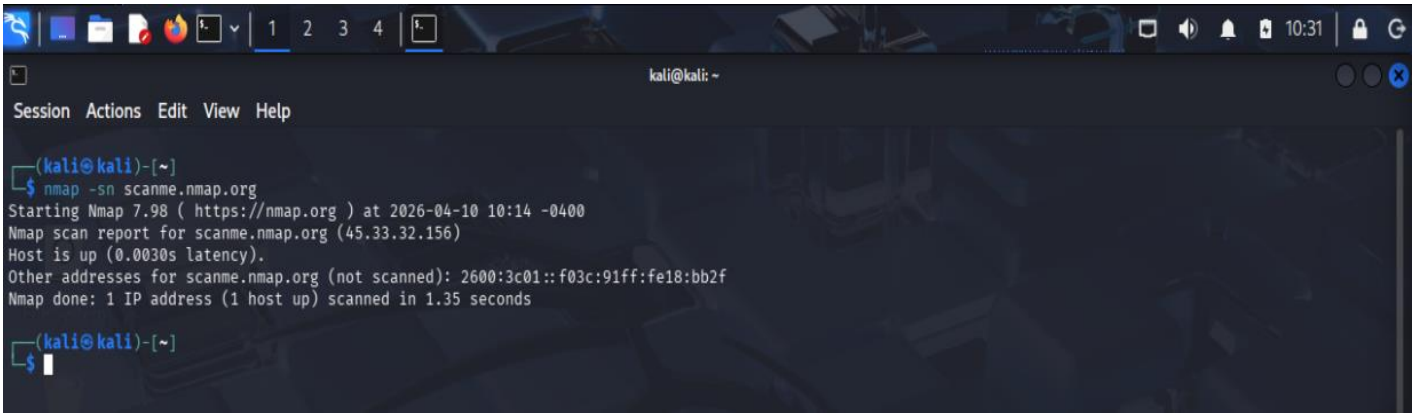
# Basic Nmap Scanning

Here we performing these scans on "**scanme.nmap.org**"

1. **Basic ping scan**

   **Command:**
   nmap -sn scanme.nmap.org




**What it does:**

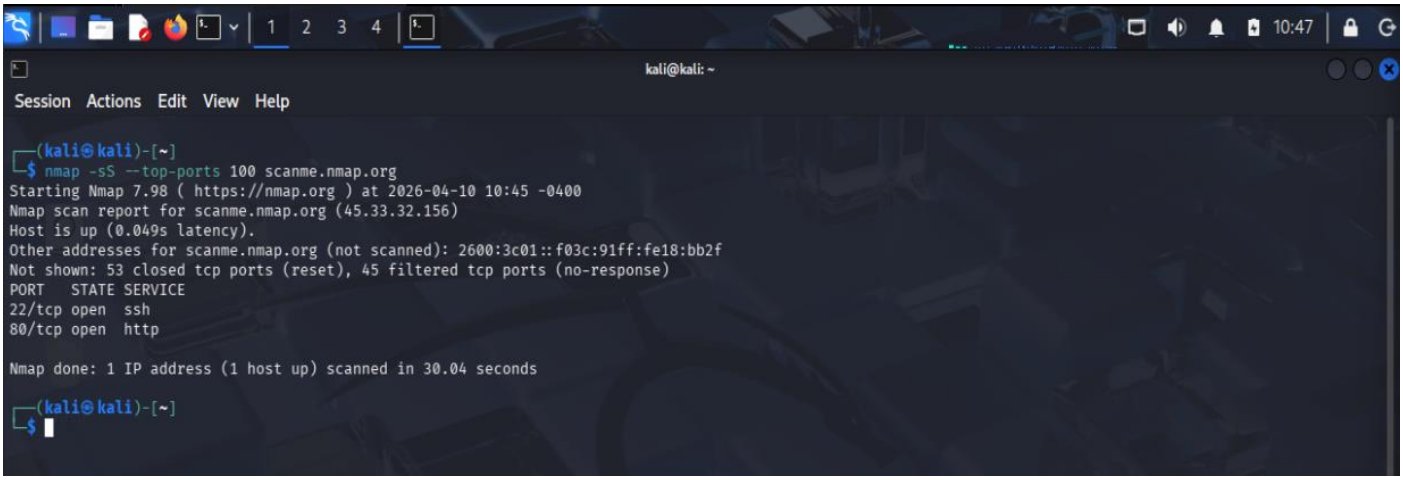
- Checks if the host is alive (reachable)
- Does not scan ports
- Uses ICMP (ping) and other probes

**Explanation:**

This scan is used to determine whether the target system is online. It does not perform port scanning, only host discovery.

2. **TCP SYN scan on top 100 ports**

   **Command:**
   nmap -sS --top-ports 100 scanme.nmap.org



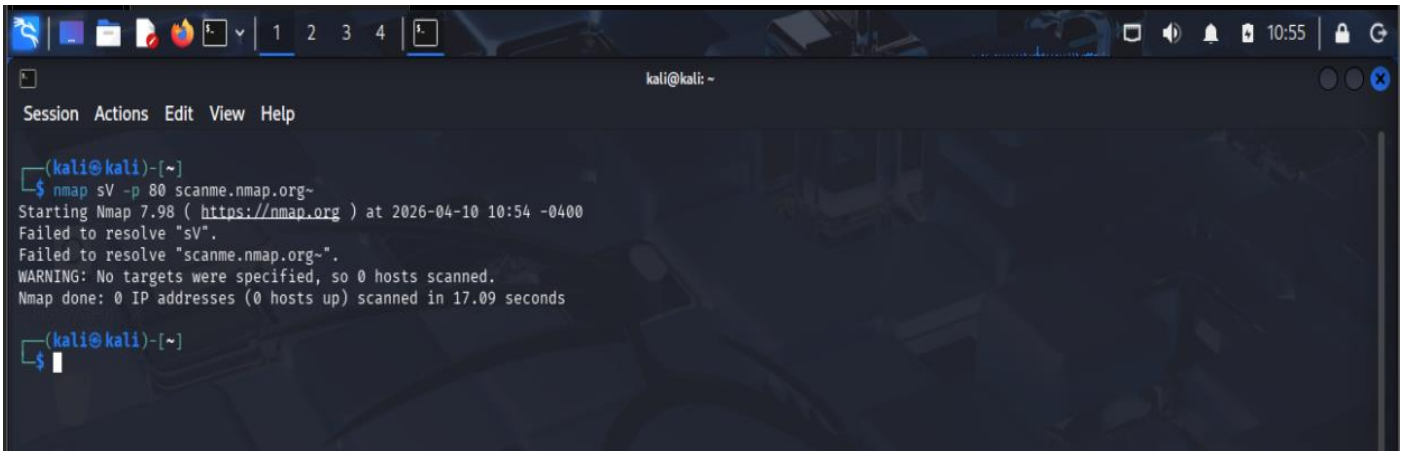

**What it does:**

- Performs a **half-open scan (stealth scan)**
- Scans most common 100 ports
- Faster and less detectable than full scan

**Explanation:**
TCP SYN scan sends SYN packets and checks responses without completing the handshake, making it faster and stealthier.

### 3. Service Version Detection

**Command:** nmap -sV -p 80 scanme.nmap.org




**What it does:**

- Detects **service and version**
- Example: Apache, Nginx, etc.

**Explanation:**

This scan identifies the service running on port 80 and its version, which helps in vulnerability assessment.

### 4. OS Detection Scan

**Command:** sudo nmap -O scanme.nmap.org

**What it does:**

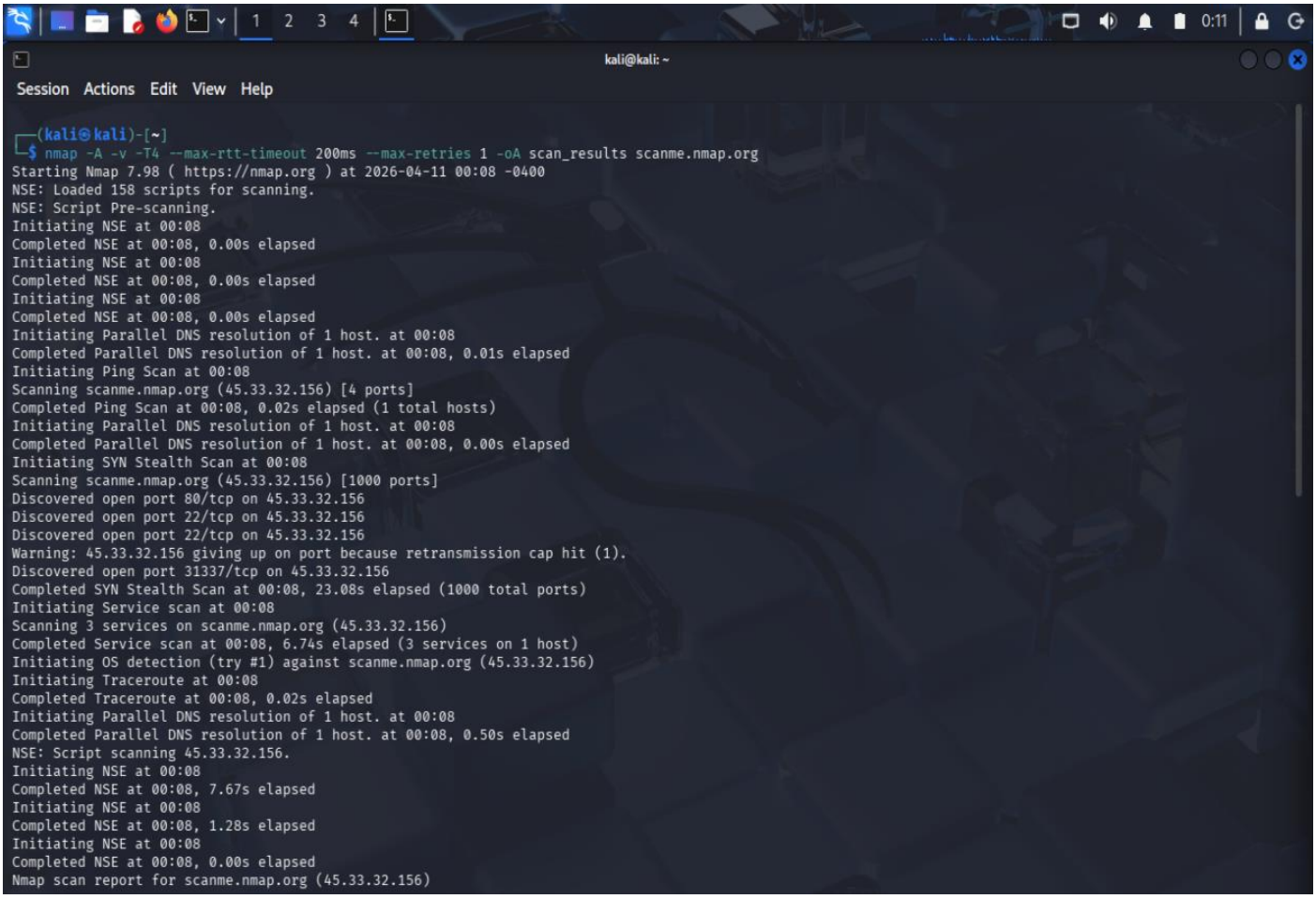
- Detects operating system
- Requires root privileges
- **Explanation:**

OS detection uses TCP/IP fingerprinting to guess the operating system of the target.

5. **Scan using default NSE scripts for each scan**

**Command:**

**nmap -A -v -T4 --max-rtt-timeout 200ms --max-retries 1 -oA scan_results scanme.nmap.org**



```
Host is up (0.023s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE    VERSION
22/tcp    open  ssh        OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http       Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
|_http-favicon: Nmap Project
| http-methods:
|_  Supported Methods: POST OPTIONS GET HEAD
31337/tcp open  tcpwrapped
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 2.4.X
OS CPE: cpe:/o:linux:linux_kernel:2.4.37
OS details: DD-WRT v24-sp2 (Linux 2.4.37)
Network Distance: 2 hops
TCP Sequence Prediction: Difficulty=254 (Good luck!)
IP ID Sequence Generation: Incremental
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)
HOP RTT     ADDRESS
1   0.41 ms 192.168.30.2
2   0.07 ms scanme.nmap.org (45.33.32.156)

NSE: Script Post-scanning.
Initiating NSE at 00:08
Completed NSE at 00:08, 0.00s elapsed
Initiating NSE at 00:08
Completed NSE at 00:08, 0.00s elapsed
Initiating NSE at 00:08
Completed NSE at 00:08, 0.00s elapsed
Read data files from: /usr/share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 43.43 seconds
           Raw packets sent: 2067 (93.172KB) | Rcvd: 955 (38.968KB)

┌──(kali㉿kali)-[~]
```

**What it does:**

This command uses Nmap to perform a **fast, detailed scan** of the target scanme.nmap.org and saves the results.

**Explanation:**

This command performs an aggressive and fast scan on the target system. It detects open ports, running services, operating system, and executes default scripts. The scan is optimized for speed using timing and retry options. The results are saved in multiple formats for further analysis.